



Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

DEPARTMENT OF SOFTWARE ENGINEERING

Computer Graphics Project

Title : CodeCraft: Visual Simulator for Understanding Algorithms Through Animation

Group-7

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Application Area: Educational Application (Interactive Learning Tool)

Objective of the Project:

The primary goal of CodeCraft is to transform the often abstract and challenging world of computer science algorithms into an intuitive and engaging visual experience. This interactive, WebGL-powered educational tool is designed to help students, particularly those studying computer science, grasp complex algorithms like sorting and pathfinding with greater ease. By dynamically translating the underlying logic of these algorithms into animated 2D visuals, CodeCraft aims to foster a deeper and more enjoyable understanding of how they work.

Brief Description of the System:

CodeCraft is a sophisticated 2D visual simulation platform built upon WebGL technology. The user interface is thoughtfully divided into two main sections: an interactive canvas where the magic happens, and a user-friendly control panel. Users can effortlessly select from a range of algorithms, such as Bubble Sort, Merge Sort, or the A* Pathfinding algorithm. Once an algorithm is chosen, the system intelligently generates a visual representation of the data it will operate on, for instance, a series of 2D rectangular bars for sorting demonstrations, or a grid of interconnected nodes for pathfinding scenarios. As the selected algorithm executes, users will witness the visual elements on the canvas dynamically change color, move, and resize in real-time, vividly illustrating the internal processes and logic of the algorithm.

Key Features and Functionalities:

- **Dynamic Data Generation:** Users have the flexibility to generate a randomized dataset (e.g., a collection of bars for sorting) or to meticulously craft a custom grid maze by simply clicking with their mouse.
- **Real-time Animation:** The system provides a step-by-step visual execution of algorithms, allowing users to observe each operation as it unfolds.
- **Interactive Controls:** A suite of intuitive controls, including play, pause, reset, and speed adjustment buttons, empowers users to manage the simulation at their own pace.

- Customizable Environment: For pathfinding algorithms, users can interactively draw "walls" or obstacles directly onto the grid using their mouse, creating unique challenges before initiating the algorithm.

Computer Graphics Concepts to be Used:

2D Graphics & Rendering: CodeCraft leverages WebGL vertex and fragment shaders to render fundamental 2D geometric shapes. This includes rectangles for representing array elements, circles or squares for grid nodes, and lines to depict connections within the data structures.

Transformations:

- Translation: This technique will be employed to animate the swapping of elements within an array or to visually represent a "search indicator" moving across a grid during algorithm execution.
- Scaling: Dynamic scaling will be used to adjust the height of bars based on their data values, or to create a subtle "pulsing" effect to highlight a node currently being evaluated by a pathfinding algorithm.
- Rotation: Rotational effects will be utilized for elements such as loading icons or to orient directional arrows, clearly indicating the movement or progression of a pathfinding algorithm.
- Animation: The requestAnimationFrame API will be central to CodeCraft's smooth visual experience, ensuring continuous updates and redraws of the WebGL canvas to create fluid, frame-by-frame transitions.
- User Interaction: The system will accurately map mouse coordinates to WebGL canvas coordinates, enabling users to precisely click and interact with specific 2D objects, such as placing a wall on a pathfinding grid.